CUSUM Change-Point Detection Analysis

Cumulative Sum Control Chart | JS Divergence + New IP Ratio + Top-K Source

Run Parameters

Dataset	txtD280_10.0_2.0_cusum_sparse.json
Timestamp	2026-02-23T14:51:31.510867
Method	CUSUM Change-Point (JS + NewIP + TopK)
Total epochs	—
Baseline epochs	— (?%)
Attack epochs	194
Malicious traffic	?%
CUSUM k-factor	0.5
CUSUM h-factor	4.0
Mode	SPARSE
Attack probability	0.3
Window size	10
Windows attacked	6

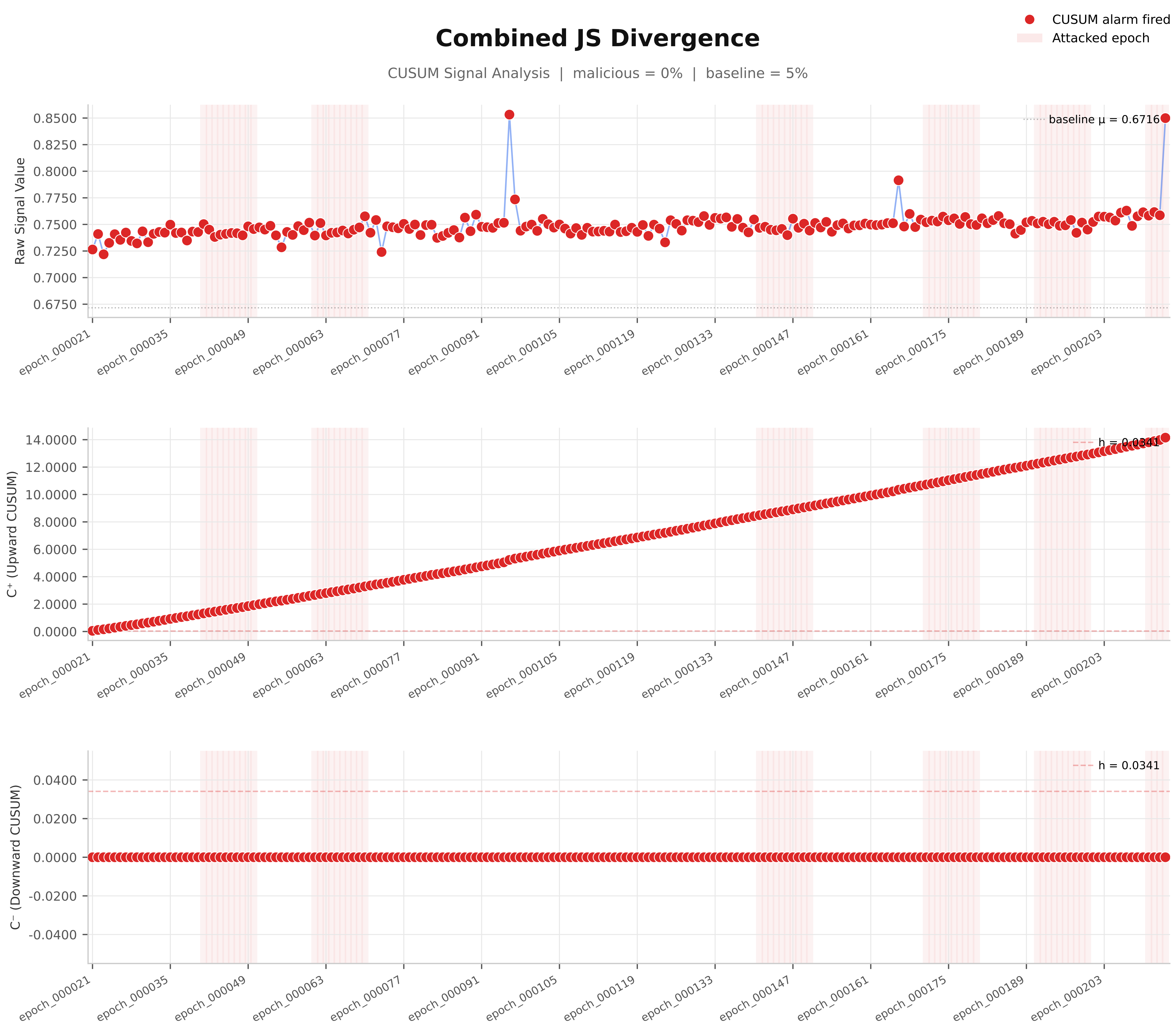
Baseline CUSUM Parameters

Combined JS	μ=0.6716 σ=0.0085 k=0.0043 h=0.0341
New-IP Ratio	μ=0.0000 σ=0.0000 k=0.0000 h=0.0000
Top-K Src	μ=0.0623 σ=0.0024 k=0.0012 h=0.0096

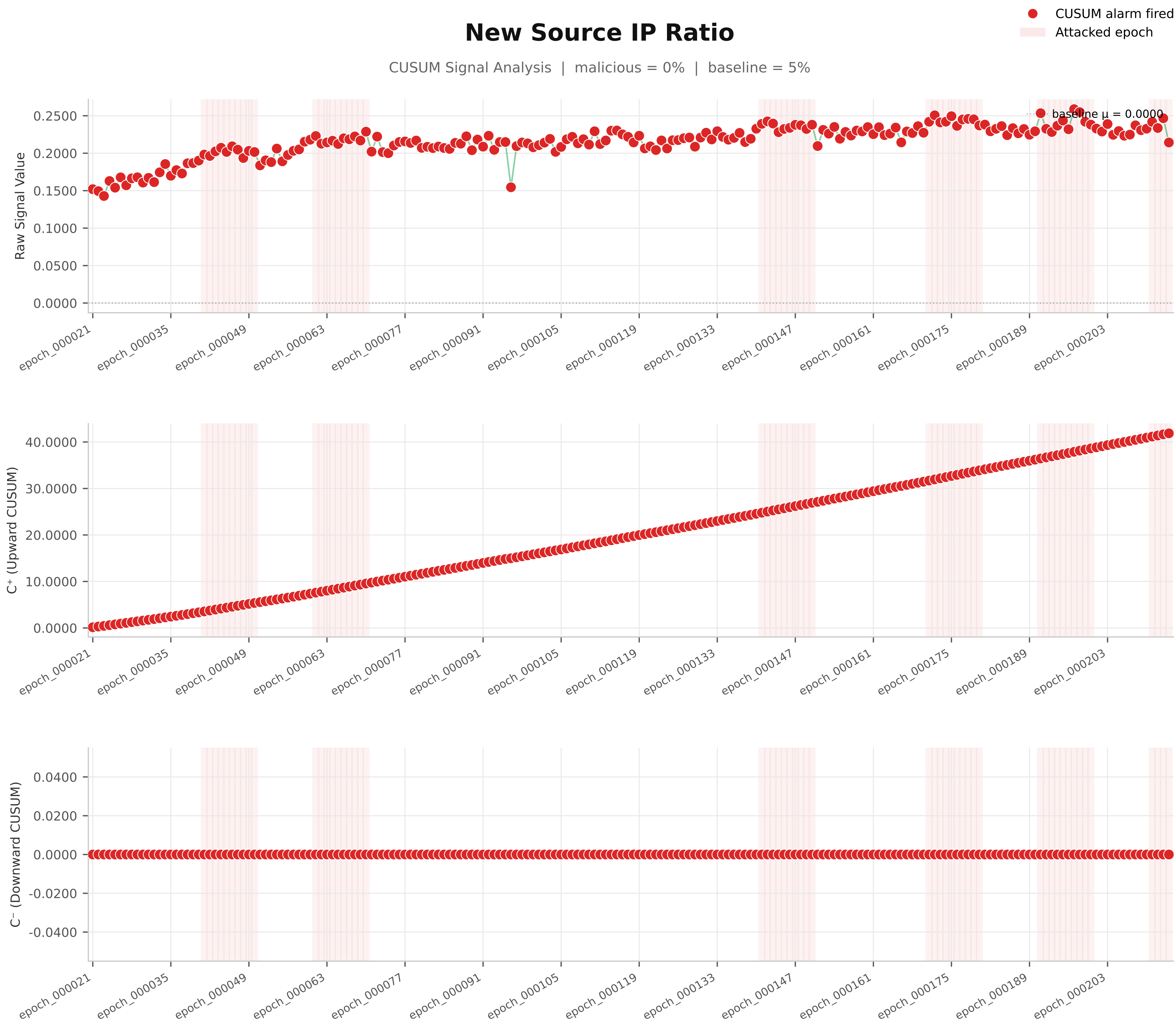
Detection Summary (Any Signal)

True Positives (TP)	**54**
False Positives (FP)	**140**
True Negatives (TN)	**0**
False Negatives (FN)	**0**
Precision	**0.278**
Recall	**1.000**
F1 Score	**0.435**
Detection Rate	**100.0%**

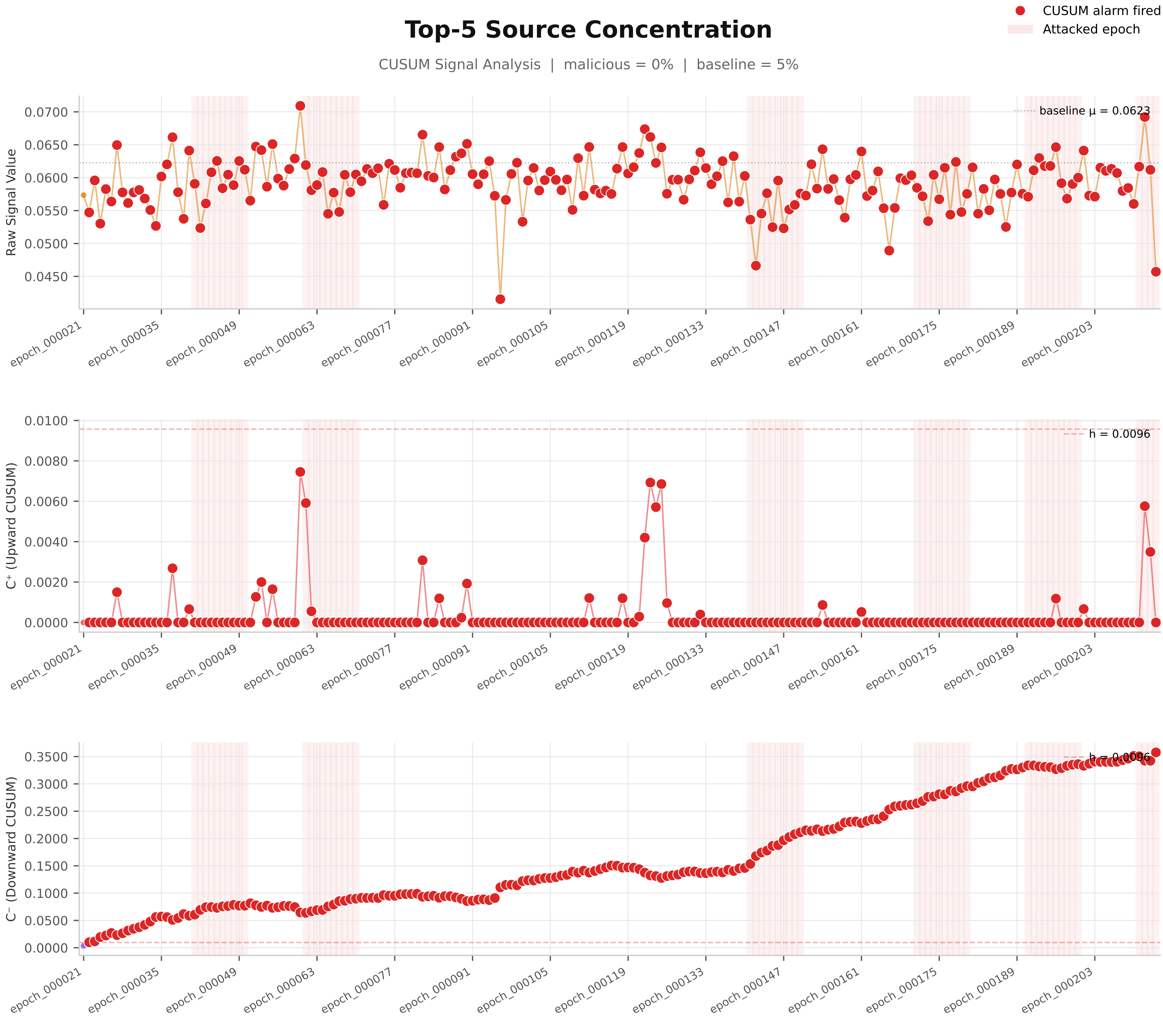
Subsequent pages: per-signal 3-panel plots (raw signal, C+, C−), alarm timeline heatmap, and detection performance table.



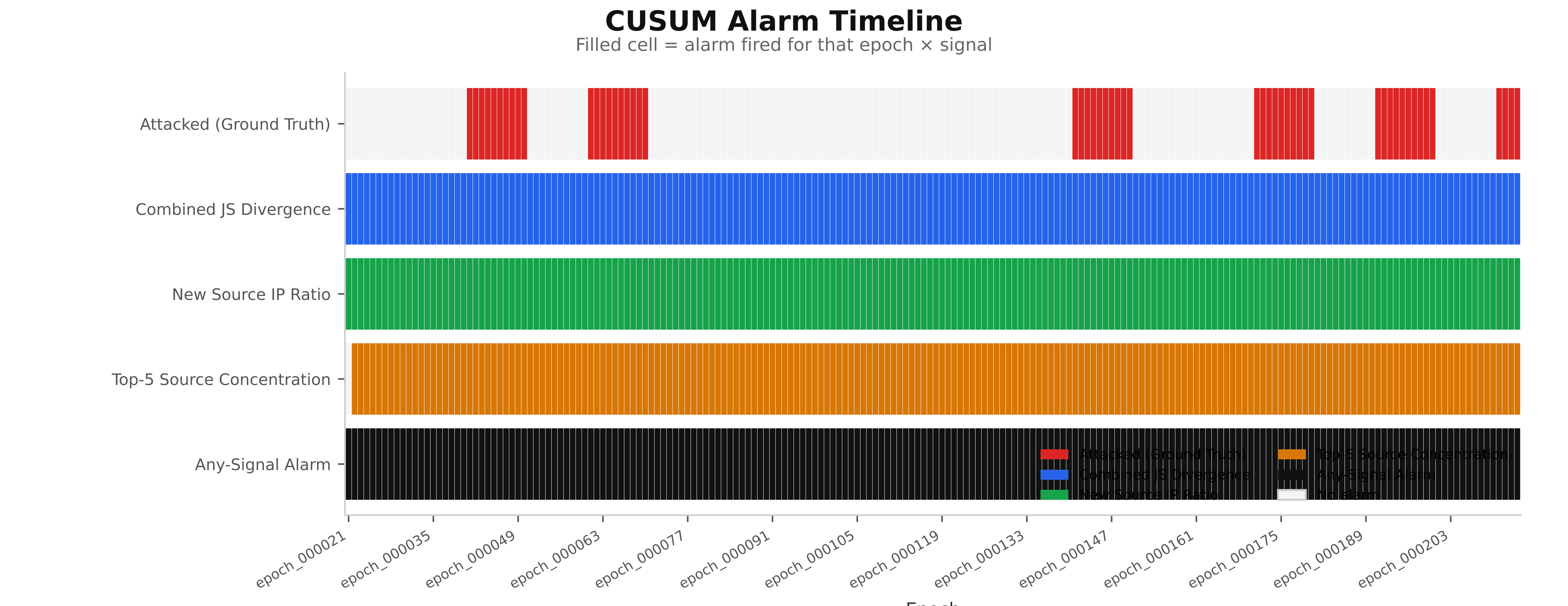
Combined JS Divergence
CUSUM Signal Analysis | malicious = 0% | baseline = 5%
CUSUM alarm fired
Attacked epoch
Raw Signal Value
baseline μ = 0.6716
C⁺ (Upward CUSUM)
h = 0.0341
C⁻ (Downward CUSUM)
h = 0.0341
epoch_000021
epoch_000035
epoch_000049
epoch_000063
epoch_000077
epoch_000091
epoch_000105
epoch_000119
epoch_000133
epoch_000147
epoch_000161
epoch_000175
epoch_000189
epoch_000203



New Source IP Ratio
CUSUM Signal Analysis | malicious = 0% | baseline = 5%
CUSUM alarm fired
Attacked epoch
baseline μ = 0.0000
Raw Signal Value
C⁺ (Upward CUSUM)
C⁻ (Downward CUSUM)
epoch_000021
epoch_000035
epoch_000049
epoch_000063
epoch_000077
epoch_000091
epoch_000105
epoch_000119
epoch_000133
epoch_000147
epoch_000161
epoch_000175
epoch_000189
epoch_000203



Top-5 Source Concentration
CUSUM Signal Analysis | malicious = 0% | baseline = 5%
CUSUM alarm fired
Attacked epoch
Raw Signal Value
baseline μ = 0.0623
C⁺ (Upward CUSUM)
h = 0.0096
C⁻ (Downward CUSUM)
h = 0.0096
0.0450
0.0500
0.0550
0.0600
0.0650
0.0700
0.0000
0.0020
0.0040
0.0060
0.0080
0.0100
0.0000
0.0500
0.1000
0.1500
0.2000
0.2500
0.3000
0.3500
epoch_000021
epoch_000035
epoch_000049
epoch_000063
epoch_000077
epoch_000091
epoch_000105
epoch_000119
epoch_000133
epoch_000147
epoch_000161
epoch_000175
epoch_000189
epoch_000203


CUSUM Alarm Timeline
Filled cell = alarm fired for that epoch × signal
Attacked (Ground Truth)
Combined JS Divergence
New Source IP Ratio
Top-5 Source Concentration
Any-Signal Alarm
No alarm
epoch_000021
epoch_000035
epoch_000049
epoch_000063
epoch_000077
epoch_000091
epoch_000105
epoch_000119
epoch_000133
epoch_000147
epoch_000161
epoch_000175
epoch_000189
epoch_000203

Detection Performance Metrics

Dataset: txtD280 | Malicious: ?% | Baseline: ?%

Metric	Value	Description
True Positives	54	Attacked epochs correctly alarmed
False Positives	140	Clean epochs incorrectly alarmed
True Negatives	0	Clean epochs correctly not alarmed
False Negatives	0	Attacked epochs missed
Precision	0.2784	TP / (TP + FP)
Recall	1.0000	TP / (TP + FN)
F1 Score	0.4355	Harmonic mean of Precision & Recall
Detection Rate	100.00%	Fraction of attacked epochs detected
False Alarm Rate	100.00%	Fraction of clean epochs with alarm